

$\frac{105}{105} \rightarrow \left(\frac{100}{100}\right)$ (2013)

Michael
Price
7-pages

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

(1) (a) Let $\Omega \neq \emptyset$. Define what it means for a collection of subsets of Ω to be:

(i) semi-algebra, (ii) algebra, (iii) σ -algebra, (iv) π -system, (v) λ -system

(b) A nonempty collection $\mathcal{M}$ is said to be a *monotone class* if it is closed under increasing unions and decreasing intersections (i.e. If $\{A_n\}$ is an increasing sequence of sets in $\mathcal{M}$ then $\bigcup_{n \geq 1} A_n \in \mathcal{M}$ and if $\{A_n\}$ is an decreasing sequence of sets in $\mathcal{M}$ then $\bigcap_{n \geq 1} A_n \in \mathcal{M}$). Prove that if $\mathcal{F}$ is a σ -algebra, it is also a

(i) algebra, (ii) π -system, (iii) λ -system, (iv) semi-algebra, (v) monotone class.

part 3 (10 × 3 = 30 points)
part 2

(2) (a) Let $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$. Define $A_n = \{j : j \geq n\}$, $n \geq 1$, $\mu(A) = |A|$ (counting measure). Show that monotone continuity from above does not hold for this $\{A_n\} \subset \mathcal{F}$.

(b) On the same $(\Omega, \mathcal{F})$ and with the same $\{A_n\} \subset \mathcal{F}$, can you specify a different measure μ' such that the monotone continuity from above does hold this $\{A_n\} \subset \mathcal{F}$?

(c) In general, under what conditions does the monotone continuity from above hold?

(d) In general, under what conditions does the monotone continuity from below hold?

all conditions (4 × 5 = 20 points)

(3) Give an example of a function f that is not $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable. — $\mathcal{B}(\mathbb{R})$

(b) Give an example of a function f that is not $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable, but $|f|$ is $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable.

(2 × 5 = 10 points)

(4) (a) Let $\Omega_1 = \mathbb{R}$ and $\mathcal{F}_1 = \{\emptyset, A, A^c, \Omega_1\}$, with $A = (0, 5]$ and let $(\Omega_2, \mathcal{F}_2) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Let $T : \Omega_1 \rightarrow \Omega_2$ be defined as

$$T(x) = \begin{cases} 5, & \text{if } x \leq 0 \\ 0, & \text{if } x > 0 \end{cases}$$

Is T $(\mathcal{F}_1, \mathcal{F}_2)$ -measurable? Justify your answer. $\wedge 0$

(b) Let $T : \mathbb{R} \rightarrow \mathbb{R}$ be defined as:

$$T(x) = e^{x^2} I_{\{x^2 > 4\}} + \tan(3x) I_{\{\cos(3x) \neq 0\}}.$$

Is T a $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable transformation? Justify your answer.

yes

- ✓ (c) Let each of $X_1, \dots, X_n$ be (real valued) random variables (defined on some $(\Omega, \mathcal{F}, P)$). Then show that $T: \Omega \rightarrow \mathbb{R}$ defined below is a $(\mathcal{F}, \mathcal{B}(\mathbb{R}))$ -measurable transformation:

$$T(\omega) = \frac{\sqrt{\frac{1}{n-1} \left(\sum_{i=1}^n X_i^2(\omega) - \left(\frac{\sum_{i=1}^n X_i(\omega)}{n} \right)^2 \right)}}{\underbrace{\left(\frac{1}{n} \sum_{i=1}^n X_i(\omega) \right)}_{\text{sample variance}}} \mathbf{I}_{\{\frac{1}{n} \sum_{i=1}^n X_i(\omega) \neq 0\}}$$

[Note: in statistics, this is the standard "plug-in estimate" for the "coefficient of variation" = σ/μ of a random variable, where μ, σ are mean and s.d of the random variable]
(3 × 10 = 30 points)

- (5) Take $F(x) = (1 - e^{-x}) \mathbf{I}_{\{x \geq 0\}}$.

how to
get
extension
(a) Using the steps in Caratheodory Extension Theorem(s), show by starting from defining a measure μ_F on a suitable semi-algebra, how one can extend it to the Lebesgue-Stieltjes measure μ_F^* on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

(b) Is such a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ unique? Why?

[Write down the steps of the extension theorem carefully and state the results used in each step]

(10 + 5 = 15 points)

add later

$$\textcircled{1} \Omega \neq \emptyset$$

(i) Let $\mathcal{F} \subset \mathcal{P}(\Omega)$. Then $\mathcal{F}$ is a semi-algebra if the following hold:

$$(i) A, B \in \mathcal{F} \Rightarrow A \cap B \in \mathcal{F}$$

$$(ii) A \in \mathcal{F} \Rightarrow \exists B_1, \dots, B_k \in \mathcal{F} \text{ w/ } 1 \leq k < \infty \text{ s.t.}$$

$$B_i \cap B_j = \emptyset \quad \forall i \neq j \text{ and}$$

$$A^c = \bigcup_{i=1}^k B_i$$

(ii) Let $\mathcal{F} \subset \mathcal{P}(\Omega)$. Then $\mathcal{F}$ is an algebra if:

$$(i) \Omega \in \mathcal{F}$$

$$(ii) A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$$

$$(iii) A, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}$$

(iii) Let $\mathcal{F} \subset \mathcal{P}(\Omega)$. Then $\mathcal{F}$ is a σ -algebra if $\mathcal{F}$ is an algebra and

$$(iv) A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$$

(iv) $\mathcal{C} \subset \mathcal{P}(\Omega)$ is a π -system if

$$(i) A, B \in \mathcal{C} \Rightarrow A \cap B \in \mathcal{C}$$

(v) $\mathcal{L} \subset \mathcal{P}(\Omega)$ is a λ -system if

$$(i) \Omega \in \mathcal{L}$$

$$(ii) A, B \in \mathcal{L} \text{ w/ } B \subset A \Rightarrow A \setminus B \in \mathcal{L}$$

$$(iii) \{A_n\}_{n \geq 1} \subset \mathcal{L} \text{ with } A_n \subset A_{n+1} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{L}$$

(b) Let $\mathcal{F}$ be a σ -algebra

(i) Then $\mathcal{F}$ is an algebra by definition since

$$(i) \Omega \in \mathcal{F}$$

$$(ii) A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$$

$$(iii) A, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}$$

(ii) Since $\mathcal{F}$ is a σ -algebra

$$A, B \in \mathcal{F} \Rightarrow A^c, B^c \in \mathcal{F}$$

$$\Rightarrow A^c \cup B^c \in \mathcal{F}$$

$$\Rightarrow (A \cap B)^c \in \mathcal{F}$$

$$\Rightarrow ((A \cap B)^c)^c \in \mathcal{F}$$

$$\Rightarrow A \cap B \in \mathcal{F}$$

Hence, $\mathcal{F}$ is a π -system.

(iii) Since $\mathcal{F}$ is a σ -algebra,

$$(i) \Omega \in \mathcal{F} \quad \checkmark$$

$$(ii) A, B \in \mathcal{F} \text{ w/ } A \supset B, \text{ then } B^c \in \mathcal{F}$$

$$\Rightarrow A \setminus B = A \cap B^c \in \mathcal{F} \quad \checkmark \quad (\sigma\text{-algebras are closed under finite unions})$$

(iii) By definition of σ -algebra, $A_n \in \mathcal{F} \forall n \geq 1$ w/

$$A_n \subset A_{n+1} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}. \quad \checkmark$$

Hence, $\mathcal{F}$ is also a λ -system.

(iv) $\mathcal{F}$ a σ -algebra

$$(i) \Rightarrow A, B \in \mathcal{F} \Rightarrow A \cap B \in \mathcal{F} \quad \checkmark$$

(ii) Let $A \in \mathcal{F}$. Then let $B_i = A^c$ and $A^c = \bigcup_{i=1}^{\infty} B_i \in \mathcal{F}$ since $A^c \in \mathcal{F}$ since $\mathcal{F}$ a σ -alg.

(v) We want to show $\mathcal{F}$ is a monotone class. Need to show that if $\{A_n\}$ is an increasing sequence of sets in $\mathcal{F}$ then

① $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$. Consider that $\bigcup_{n=1}^{\infty} A_n$ is a countable union of sets in $\mathcal{F}$, so by the definition of a σ -algebra, it must be that

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}.$$

We also need to show that if $\{A_n\}$ is a decreasing sequence of sets in $\mathcal{F}$, then $\bigcap_{n=1}^{\infty} A_n \in \mathcal{F}$. Consider that

$$A_n \in \mathcal{F} \Rightarrow A_n^c \in \mathcal{F}$$

$$\Rightarrow \bigcup_{n=1}^{\infty} A_n^c \in \mathcal{F}$$

$$\Rightarrow \left(\bigcap_{n=1}^{\infty} A_n \right)^c \in \mathcal{F}$$

$$\Rightarrow \left(\left(\bigcap_{n=1}^{\infty} A_n \right)^c \right)^c \in \mathcal{F}$$

$$\Rightarrow \bigcap_{n=1}^{\infty} A_n \in \mathcal{F}. \quad \checkmark$$

Hence, $\mathcal{F}$ is a monotone class.

②^(a) $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$, $A_n = \{j : j \geq n\}$, $n \geq 1$, $\mu(A) = |A|$.

Want to show that monotone continuity from above does not hold for $\{A_n\} \subset \mathcal{F}$. In order for MCT to hold:

$$\lim_{n \rightarrow \infty} \mu(A_n) = \mu \left(\bigcap_{n=1}^{\infty} A_n \right)$$

① In this case,

$$\mu(A_n) = |A_n| = |\mathbb{N}| = \infty$$

$$\Rightarrow \lim_{n \rightarrow \infty} \mu(A_n) = \infty.$$

Additionally,

$$\begin{aligned}\bigcap_{n=1}^{\infty} A_n &= \left(\left(\bigcap_{n=1}^{\infty} A_n \right)^c \right)^c \\ &= \left(\bigcup_{n=1}^{\infty} A_n^c \right)^c \\ &= \left(\bigcup_{n=1}^{\infty} \{1, 2, \dots, n\} \right)^c \\ &= (\mathbb{N})^c \\ &= \emptyset\end{aligned}$$

$$\Rightarrow \mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \mu(\emptyset) = |\emptyset| = 0$$

Hence,

$$\lim_{n \rightarrow \infty} \mu(A_n) = \infty \neq 0 = \mu\left(\bigcap_{n=1}^{\infty} A_n\right).$$

(b) Define $\mu'(A) = 0 \quad \forall A \in \mathcal{F}$. Then

$$(i) \mu'(\emptyset) = 0$$

$$(ii) \mu'(A) \in [0, \infty] \quad \forall A \in \mathcal{F}$$

(iii) For any sequence of disjoint sets $\{A_n\}_{n=1}^{\infty} \in \mathcal{F}$ with $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$ (true since $\mathcal{F}$ a σ -algebra),

then

$$\mu'\left(\bigcup_{n=1}^{\infty} A_n\right) = 0 = \sum_{n=1}^{\infty} \mu'(A_n) \quad \checkmark$$

Hence, μ' is a measure, and for $\{A_n\}_{n=1}^{\infty} \subset \mathcal{F}$ with $A_n \supset A_{n+1}$,

$$\begin{aligned}\lim_{n \rightarrow \infty} \mu'(A_n) &= 0 \quad \text{and} \quad \bigcap_{n=1}^{\infty} A_n = \emptyset, \text{ so} \\ \mu'\left(\bigcap_{n=1}^{\infty} A_n\right) &= 0 \Rightarrow \lim_{n \rightarrow \infty} \mu'(A_n) = \mu'\left(\bigcap_{n=1}^{\infty} A_n\right). \checkmark\end{aligned}$$

(c) In order for MCFA to hold, there must be some $n \in \mathbb{N}$ s.t.

$$\mu(A_{n_0}) < \infty.$$

(d) Holds under any conditions in which there is a sequence of sets with $\{A_n\}_{n=1}^\infty$ s.t. $A_n \subset A_{n+1}$ and μ a measure.

5 $F(x) = (1 - e^{-x}) \mathbb{I}_{\{x \geq 0\}}$

(a) Let

$$\mathcal{C} = \{(a, b] : -\infty \leq a \leq b < \infty\} \cup \{(a, \infty) : -\infty \leq a < \infty\},$$

and it can be shown that $\mathcal{C}$ is a semi-algebra.
($\mathcal{C} \subset \mathcal{P}(\mathbb{R})$).

Consider $F(x)$. WTS $F(x)$ is a non-decreasing function. Consider that

$$\begin{aligned} \frac{d}{dx} F(x) &= \frac{d}{dx} (1 - e^{-x}) \\ &= -e^{-x}(-1) \\ &= e^{-x} \\ &= \frac{1}{e^x} > 0 \quad \forall x \geq 0 \end{aligned}$$

Hence, $F(x)$ is non-decreasing. ✓

Then we can define a function μ_F as follows.

$$\begin{aligned} \mu_F((a, b]) &= F(b+) - F(a+) \\ &= F(b) - F(a) \quad (\text{since } F \text{ is continuous for } x \geq 0) \longrightarrow \end{aligned}$$

$$= (1 - e^{-b}) - (1 - e^{-a})$$

$$= 1 - e^{-b} - 1 + e^{-a}$$

$$= \frac{1}{e^a} - \frac{1}{e^b}$$

$$\mu_F((a, \infty)) = F(\infty) - F(a+)$$

$$= F(\infty) - F(a) \quad (\text{since } F \text{ is continuous for } x \geq 0)$$

$$= (1 - e^{-\infty}) - (1 - e^{-a})$$

$$= 1 - 0 - 1 + e^{-a}$$

$$= \frac{1}{e^a}$$

Then μ_F can be shown to be a measure.

Caratheodory's extension theorem says that for a semi-algebra $\mathcal{C}$ and a measure μ_F defined on $\mathcal{C}$, let us define a set function

$$\mu_F^* : \mathcal{C} \rightarrow \overline{\mathbb{R}}^+ \equiv [0, \infty] \text{ as}$$

$$\mu_F^*(A) \equiv \inf \left\{ \sum_{n=1}^{\infty} \mu_F(A_n) : \{A_n\}_{n=1}^{\infty} \subset \mathcal{C} \text{ and } A \subset \bigcup_{n=1}^{\infty} A_n \right\}.$$

Then

(i) μ_F^* is an outer measure

(ii) μ_F^* restricted to $\mathcal{M} = \{A : A \text{ is } \mu^* \text{-measurable}\}$ is a measure $= \mu$

(iii) $\mathcal{C} \subset \mathcal{M}$.

This μ_F^ is the Lebesgue-Stieltjes measure on μ_F^* and

$(\mathbb{R}, \mathcal{M}, \mu_F^*)$ is the Caratheodory extension of μ .
(and is known as the Lebesgue-Stieltjes measure space)

Since $\sigma(\mathcal{C}) \subset \mathcal{M}$, $(\Omega, \sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R}, \mu_F^*))$ is also a measure space and μ_F^* is called the Lebesgue-Stieltjes measure.

(b) The extension theorem says that for two finite measures μ_1 and μ_2 on a measurable space $(\Omega, \mathcal{F})$. Let $\mathcal{C} \subset \mathcal{F}$ be a π -class s.t.

$$\sigma(\mathcal{C}) = \mathcal{F}. \text{ If}$$

$$\mu_1(C) = \mu_2(C)$$

for all $C \in \mathcal{C}$, and

$$\mu_1(\Omega) = \mu_2(\Omega),$$

then

$$\mu_1(A) = \mu_2(A)$$

for all $A \in \mathcal{F}$.

We have that $\mathcal{C}$ in our case is a semi-algebra, so it will be a π -system. Additionally, it can be shown that $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R})$. Thus, our measurable space in this case is $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Note that μ^* is finite on $\mathcal{C}$ since $\mu^* = \mu$ on $\mathcal{C}$

and

$$\mu((-\infty, \infty)) = F(\infty) - F(-\infty) = 1 - 0 = 1 < \infty.$$

Hence, $\mu^*(\Omega) < \infty$ on $\mathcal{C}$.

→

Then by the extension theorem, any two
finite measures μ_1 and μ_2 on $\mathcal{C}$ that are
equal will be equal on the bigger class $(\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C}))$,
so this extension is unique.

3 (a) Let A be a non-Borel measurable set. Then $f = I_A$ is not $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ -measurable.

(b) Let $f = I_A - I_{A^c}$. Then

$$f(x) = \begin{cases} 1 & \text{if } x \in A \\ -1 & \text{if } x \notin A \end{cases}$$

and f is not $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ -measurable. However, $|f| = 1$, and constant functions are $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ -measurable.

4 (a) $\Omega_1 = \mathbb{R}$, $\mathcal{F}_1 = \{\emptyset, (0, 5], (-\infty, 0] \cup (5, \infty)\}$.

$\Omega_2 = \mathbb{R}$ $\mathcal{F}_2 = \mathcal{B}(\mathbb{R})$

Let $A \in \mathcal{F}_2$. Then

$$T^{-1}(A) = \begin{cases} \emptyset & \text{if } 0 \notin A, 5 \notin A \\ (-\infty, 0] & \text{if } 0 \notin A, 5 \in A \\ (0, \infty) & \text{if } 0 \in A, 5 \notin A \\ \mathbb{R} & \text{if } 0 \in A, 5 \in A. \end{cases}$$

Note that $(0, \infty) \notin \mathcal{F}_1$, so T is not $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ -measurable.

(b) $T(x) = e^{x^2} I_{\{x^2 > 4\}} + \tan(3x) I_{\{\cos(3x) \neq 0\}}$

Here, $T(x)$ is composed of pointwise addition and multiplication of measurable functions, so $T(x)$ is $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ -measurable.

(c) Here again, $T(\omega)$ is composed of pointwise addition, pointwise multiplication, and scalar multiplication of measurable functions. Thus,

$T(\omega)$ is $(\mathcal{F}, \mathcal{B}(\mathbb{R}))$ -measurable.